



MIS 7397 | Selected Topics in Management Information Systems

Section: 27425 | Face-to-Face

Prerequisite(s): Graduate standing and approval of chair or program director. May be repeated when topics vary.

Instructor Information

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Office Hours

Will be announced on Canvas.

General Course Information

MIS 7397

Selected Topics in Management Information Systems Course Title:

Introduction to Artificial Intelligence for Business

Fall 2026

Credit Hours: 3

Course Level: Graduate

Instruction Mode: Face-to-Face

Section: 27245

Course Objectives and Student Learning Outcomes

This course prepares graduate business students to understand, evaluate, design, and responsibly lead AI-enabled initiatives. The course emphasizes business problem framing, technical intuition, model evaluation, implementation, economics, governance, cybersecurity, and organizational impact.

Upon successful completion of the course, students will be able to:

1. Distinguish artificial intelligence from conventional analytics, business rules, automation, machine learning, deep learning, and generative AI.
2. Translate an organizational need into a clearly scoped AI use case, decision, workflow, user, and measurable business objective.
3. Assess whether an AI approach is appropriate and identify when a non-AI alternative is preferable.
4. Evaluate data readiness, including data quality, representativeness, privacy, ownership, access, and potential bias.
5. Explain the intuition behind supervised learning, unsupervised learning, recommendation systems, neural networks, language models, computer vision, retrieval-augmented generation, and AI agents.
6. Interpret model and system performance using metrics that reflect business consequences, including false positives, false negatives, cost, latency, reliability, and human review.
7. Use approved no-code or low-code tools to construct and evaluate a basic AI prototype or workflow.
8. Analyze the strategic and economic value of an AI initiative, including feasibility, return on investment, build-versus-buy considerations, and implementation tradeoffs.
9. Identify ethical, legal, privacy, security, and societal risks and recommend appropriate governance, monitoring, and human-oversight controls.
10. Communicate AI recommendations clearly to executive, operational, and technical audiences using evidence, professional writing, and effective presentation design.

Required Instructional Materials

Required Textbook and Courseware:

Corinne Hoisington and Mark Ciampa, Introduction to Artificial Intelligence: A Business Perspective, 1st Edition, Cengage, 2026.

MindTap for Hoisington/Ciampa's Introduction to Artificial Intelligence: A Business Perspective, 1-term Instant Access, ISBN 9798214024332, is required and will be used for selected readings, concept checks, homework, and publisher-supported learning activities. MindTap will be integrated with Canvas when available.

Recommended Supplemental Text, Optional:

Kamales Lardi, Artificial Intelligence for Business: Harness AI for Value, Growth and Innovation, 1st Edition, Kogan Page, 2025, ISBN 9781398618008. This text is not required for purchase and is recommended for students who want additional depth in AI strategy, organizational transformation, responsible implementation, leadership, and enterprise adoption.

Additional Required Materials and Technology Additional readings, cases, research summaries, policy documents, technical explainers, and current practitioner articles will be posted in Canvas. Students will also need regular access to a laptop with a modern web browser, Microsoft Office or equivalent productivity software, and university-approved or instructor-approved AI and analytics tools. No prior programming experience is required.

Course Schedule, Assignments, and Assessments

Week	Topic	Major Assignment / Assessment
1	AI, Business, and Competitive Transformation	AI opportunity observation
2	From Business Problem to AI Problem	Capstone opportunity proposal

3	Data Foundations and the AI Factory	Data readiness exercise
4	Supervised Machine Learning for Business	Applied Lab 1
5	Unsupervised Learning, Recommendations, and Personalization	Executive Memo 1
6	Model Evaluation and Experimentation	Applied Lab 2
7	Deep Learning, Natural Language, and Computer Vision	Current AI briefing
8	Generative AI and Large Language Models	Midterm AI decision case
9	Retrieval-Augmented Generation, Agents, and Intelligent Workflows	Capstone prototype plan
10	From Prototype to Production	Vendor evaluation memo
11	AI Across Business Functions and Industries	Applied Lab 3 / case briefing
12	AI Strategy, Economics, and Competitive Advantage	Capstone feasibility and value analysis
13	Responsible AI, Law, Privacy, Security, and Governance	Capstone governance plan
14	The AI-Powered Organization	Capstone draft / peer review
15	Capstone Presentations and the Future of AI	Final report and presentation
Final Exam Period	Course Integration and Reflection	Final assessment or capstone-related evaluation as specified in Canvas

Discussion and Lecture Topics

Course discussions and lectures will cover the following major topics:

- Artificial intelligence, business transformation, and competitive advantage

- Differences among AI, analytics, automation, machine learning, deep learning, and generative AI
- Business problem framing and identifying appropriate AI use cases
- Data foundations, data quality, labeling, privacy, ownership, and representativeness
- Supervised machine learning for classification and regression
- Unsupervised learning, clustering, segmentation, recommendations, and anomaly detection
- Model evaluation, including accuracy, precision, recall, error costs, overfitting, and experimentation
- Deep learning and neural-network concepts
- Natural language processing and computer vision applications
- Generative AI, large language models, prompting, hallucination, multimodality, and verification
- Retrieval-augmented generation and enterprise knowledge grounding
- AI agents, intelligent workflows, orchestration, and human-in-the-loop design
- AI deployment, MLOps, LLMOps, monitoring, reliability, latency, and cost
- AI cybersecurity, including prompt injection and system vulnerabilities
- AI applications across marketing, finance, accounting, operations, supply chain, human resources, customer service, cybersecurity, healthcare, and product development
- AI strategy, economics, ROI, build-versus-buy decisions, and competitive advantage
- Responsible AI, bias, fairness, explainability, transparency, accountability, privacy, security, and governance
- AI organizational maturity, workforce redesign, change management, leadership, and culture
- Emerging AI technologies, the future of work, and leadership responsibilities

Topics may be adjusted during the semester to incorporate significant developments in artificial intelligence and relevant business applications.

Grading Rubrics and Weights

Concept Checks and Reading Quizzes	10%
Applied AI Laboratories	20%
Business Cases and Executive Memos	15%
Midterm AI Decision Case	15%
AI Opportunity-to-Deployment Capstone	30%
Participation and Current AI Briefings	10%
Total	100%

The capstone project includes multiple milestones covering opportunity identification, data and feasibility analysis, prototyping, evaluation and governance, the final executive report, and the executive presentation. Detailed grading criteria and rubrics for assignments and assessments will be provided through Canvas.

Course Policies and Procedures

Attendance, Preparation, and Participation:

Regular attendance and active participation are expected because many course activities involve live demonstrations, business cases, simulations, laboratories, and team workshops. Students should arrive prepared, complete assigned readings, and participate professionally. Attendance alone does not earn full participation credit; preparation, contribution, and quality of engagement are also considered. Excused absences will be handled in accordance with University policy and may require an alternative learning activity. Unexcused missed in-class work is normally not eligible for make-up credit.

Communication:

Course announcements, assignment instructions, grades, and schedule updates will be communicated through Canvas. Students are responsible for checking

Canvas and their UH email regularly. Email messages should include MIS 7397 in the subject line and clearly identify the question or requested action. The instructor will generally respond within two business days, excluding weekends, holidays, and university closures.

Due Dates, Late Work, and Extensions:

Assignments are due at the time listed in Canvas. Unless an assignment states otherwise, work submitted within 48 hours after the deadline may be accepted with a 10% deduction per calendar day or portion of a day. Work more than 48 hours late will normally receive no credit. Approved accommodations, documented excused absences, or an extension granted before the deadline supersede this rule. Team milestones may have stricter deadlines because delays can affect other students. Technical problems should be documented promptly and do not automatically excuse late work.

Collaboration and Teamwork:

Collaboration is encouraged when an assignment is designated as collaborative. Students may discuss concepts and help one another understand tools, but individual submissions must reflect each student's own analysis and writing. Sharing completed individual work, copying another team's deliverable, or submitting work produced by a person outside the authorized team is prohibited. Teams are expected to establish roles, communication norms, deadlines, and a process for resolving disagreements.

Regrading and Feedback:

Students should review feedback before requesting reconsideration. A regrade request must be submitted in writing within seven calendar days of the grade being posted and must identify the specific rubric criterion or factual issue. A regrade may result in the score increasing, remaining unchanged, or decreasing if the full submission is re-evaluated.

Technology and Continuity:

Students are responsible for maintaining access to Canvas, required files, and course tools. Important work should be backed up regularly. If a platform becomes unavailable, the instructor may substitute an equivalent tool, dataset, or

activity. Students should report accessibility, account, or integration problems early rather than waiting until a deadline.

Use of Artificial Intelligence Tools:

Because responsible use of AI is a central learning objective of this course, AI tools will be permitted or required in many activities. Permission is assignment-specific. Each assignment will identify the authorized level of AI use. Using AI beyond the authorized level, concealing material AI assistance, or submitting unverified AI-generated content as original work may constitute unauthorized assistance or academic dishonesty.

When AI use is permitted, students may be required to disclose the tool or model used, the purpose of use, important prompts or interactions, which outputs materially influenced the submission, how information was verified, and what the student independently revised or contributed.

Students remain responsible for every claim, citation, calculation, recommendation, and artifact they submit. Fabricated sources, data, quotations, citations, or results are unacceptable. Students may not enter confidential, proprietary, personally identifiable, health, financial, regulated, client, employer, or research-participant data into an AI service without authorization and an approved secure environment.

Data Security and Privacy:

Course projects must use public, synthetic, de-identified, licensed, or instructor-approved data. Students should document data sources and consider privacy and security throughout the project lifecycle. Sensitive information should not be entered into tools when data-retention or training practices are unclear.

Academic Honesty:

All course work must comply with the University of Houston Academic Honesty Policy. Potential violations include plagiarism, unauthorized assistance, falsification or fabrication, unauthorized collaboration, misrepresentation of AI use, and submission of work completed by another person or service.

Recording and Distribution of Course Materials:

Students may not record, livestream, photograph, distribute, post, sell, or upload class sessions or course materials without prior written authorization, except as permitted by an approved accommodation. Authorized recordings and materials are for personal educational use only and may not be shared.

Changes to the Course:

The instructor may make reasonable changes to topics, readings, tools, assignments, due dates, or procedures in response to significant developments in AI, platform availability, University requirements, or student learning needs. Material changes will be announced through Canvas.

University Policies and Student Support Resources

Mental Health and Wellness Resources

The University of Houston has a number of resources to support students' mental health and overall wellness, including [CoogsCARE](#) and the [UH Go App](#). [UH Counseling and Psychological Services \(CAPS\)](#) offers 24/7 mental health support for all students, addressing various concerns like stress, college adjustment and sadness. CAPS provides individual and couples counseling, group therapy, workshops and connections to other support services on and off- campus. For assistance visit uh.edu/caps, call 713-743-5454, or visit a [Let's Talk](#) location in-person or virtually. Let's Talk are daily, informal confidential consultations with CAPS therapists where no appointment or paperwork is needed.

Need Support Now? If you or someone you know is struggling or in crisis, help is available. Call CAPS crisis support 24/7 at 713-743-5454, or the National Suicide and Crisis Lifeline: call or text 988, or chat 988lifeline.org.

Title IX/Sexual Misconduct

Per the UHS Sexual Misconduct Policy, your instructor is a “responsible employee” for reporting purposes under Title IX regulations and state law and must report incidents of sexual misconduct (sexual harassment, non-consensual sexual contact, sexual assault, sexual exploitation, sexual intimidation, intimate partner violence, or stalking) about which they become aware to the Title IX

office (known at UH as the Equal Opportunity Services office or "EOS"). Please know there are places on campus where you can make a report in confidence. You can find more information about resources on the UH [Title IX/Sexual Misconduct Resources page](#). Please note that you may also report concerns of discrimination based on your protected class identity to EOS.

Reasonable Academic Adjustments/Auxiliary Aids

The University of Houston is committed to providing an academic environment and educational programs that are accessible for its students. Any student with a disability who is experiencing barriers to learning, assessment or participation is encouraged to contact the Justin Dart, Jr. Student Accessibility Center (Dart Center) to learn more about academic accommodations and support that may be available to them. Students seeking academic accommodations will need to register with the Dart Center as soon as possible to ensure timely implementation of approved accommodations. Please contact the Dart Center by visiting the website: <https://uh.edu/accessibility/> calling (713) 743-5400, or emailing jdccenter@Central.UH.EDU.

The [Student Health Center](#) offers a Psychiatry Clinic for enrolled UH students. Call 713-743-5149 during clinic hours, Monday through Friday 8 a.m. - 4:30 p.m. to schedule an appointment.

The [A.D. Bruce Religion Center](#) offers spiritual support and a variety of programs centered on well-being.

The [Center for Student Advocacy and Community \(CSAC\)](#) is where you can go if you need help but don't know where to start. CSAC is a "home away from home" and serves as a [resource hub](#) to help you get the resources needed to support academic and personal success. Through our [Cougar Cupboard](#), all students can get up to 30 lbs of FREE groceries a week. Additionally, we provide 1:1 appointments to get you connected to on- and off-campus resources related to essential needs, safety and advocacy, and more. The [Cougar Closet](#) is a registered student organization advised by our office and offers free clothes to students so that all Coogs can feel good in their fit. We also host a series of cultural and community-based events that foster social connection and helps the cougar

community come closer together. Visit the CSAC homepage or follow us on Instagram: @uh_CSAC and @uhcupbrd. YOU belong here.

Cougar Parent and Family Engagement Office

This office, under the Dean of Students Office (<https://www.uh.edu/dos/>), coordinates resources for Pregnant and Parenting Students (<https://www.uh.edu/dos/parenting-students/>), Parents and Family members of UH students (<https://www.uh.edu/dos/parents/>), and Sexual Misconduct Support Services (<https://www.uh.edu/dos/sexual-misconduct-support-services/>). A description of available support and resources are provided on the respective web pages. The Cougar Parent and Family Engagement Office is located in Student Center South, room B12.

Academic Honesty Policy

High ethical standards are critical to the integrity of any institution, and bear directly on the ultimate value of conferred degrees. All UH community members are expected to contribute to an atmosphere of the highest possible ethical standards. Maintaining such an atmosphere requires that any instances of academic dishonesty be recognized and addressed. The [UH Academic Honesty Policy](#) is designed to handle those instances with fairness to all parties involved: the students, the instructors, and the University itself. All students and faculty of the University of Houston are responsible for being familiar with this policy.

Excused Absence Policy

Regular class attendance, participation, and engagement in coursework are important contributors to student success. Absences may be excused as provided in the University of Houston [Undergraduate Excused Absence Policy](#) and [Graduate Excused Absence Policy](#) for reasons including medical illness of student or close relative, death of a close family member, legal or government proceeding that a student is obligated to attend, recognized professional and educational activities where the student is presenting, and University-sponsored activity or athletic competition. Under these policies, students with excused absences will be provided with an opportunity to make up any quiz, exam or other work that contributes to the course grade or a satisfactory alternative. Please read the full

policy for details regarding reasons for excused absences, the approval process, and extended absences. Additional policies address absences related to [military service](#), [religious holy days](#), [pregnancy and related conditions](#), and [disability](#).

Recording of Class

Students may not record all or part of class, livestream all or part of class, or make/distribute screen captures, without advanced written consent of the instructor. If you have or think you may have a disability such that you need to record class-related activities, please contact the [Justin Dart, Jr. Student Accessibility Center](#). If you have an accommodation to record class-related activities, those recordings may not be shared with any other student, whether in this course or not, or with any other person or on any other platform. Classes may be recorded by the instructor. Students may use instructor's recordings for their own studying and notetaking. Instructor's recordings are not authorized to be shared with anyone without the prior written approval of the instructor. Failure to comply with requirements regarding recordings will result in a disciplinary referral to the Dean of Students Office and may result in disciplinary action.